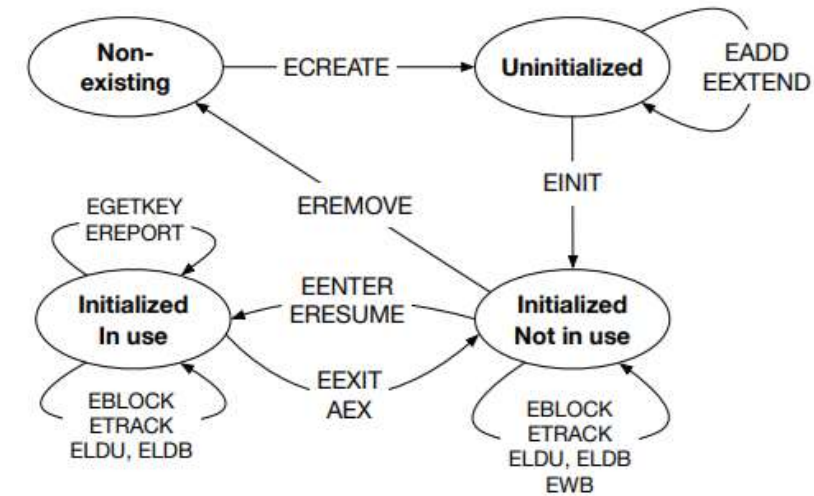


# Enclave Lifecycle



## Creation

ECREATE instruction loads a new SECS into a free EPC page, establishing the enclave's foundation.

## Initialization

EINIT instruction finalizes the enclave, enabling execution and preventing further loading operations.

**1****2****3****4**

## Loading

EADD and EEXTEND instructions load initial code and data while updating enclave measurements.

## Teardown

EREMOVE instruction frees EPC pages, completing the enclave lifecycle and releasing resources.

# Entry and Exit Mechanisms

1

## Synchronous Entry

EENTER instruction performs controlled jump to predefined addresses, preventing security bypass attacks.

2

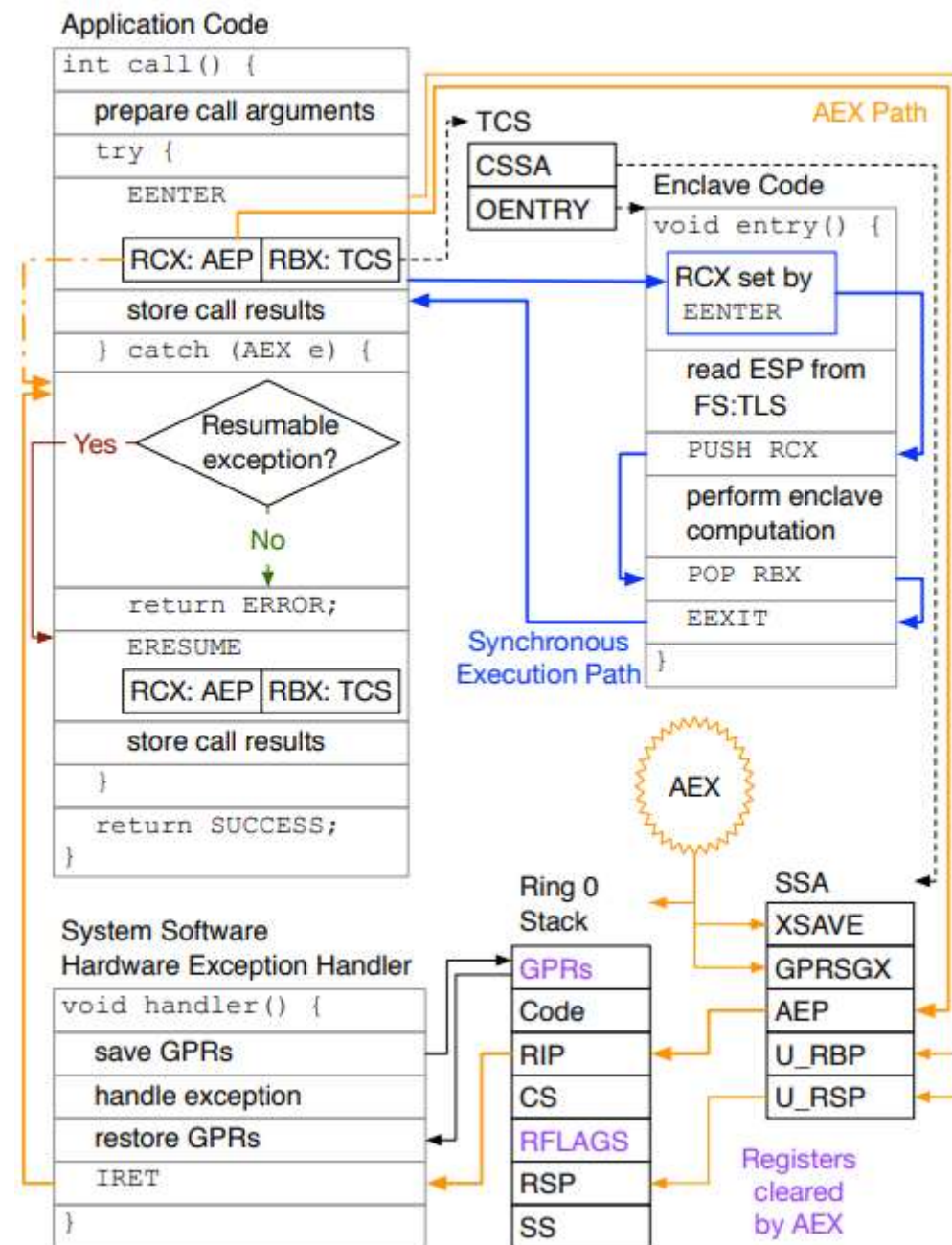
## Asynchronous Exit

Hardware exceptions trigger AEX instruction, saving context before invoking exception handler.

3

## Resumption

ERESUME instruction restores enclave execution after interrupt handling completes.



# Local Attestation



## Establish Channel

Prover Enclave obtains Verifier Enclave's measurement through a secure communication channel.



## Generate Report

Prover Enclave invokes EREPORT to create a REPORT structure and transmits it to the Verifier.



## Verify Report

Verifier Enclave retrieves the Report Key using EGETKEY and computes MAC to verify the REPORT's authenticity.

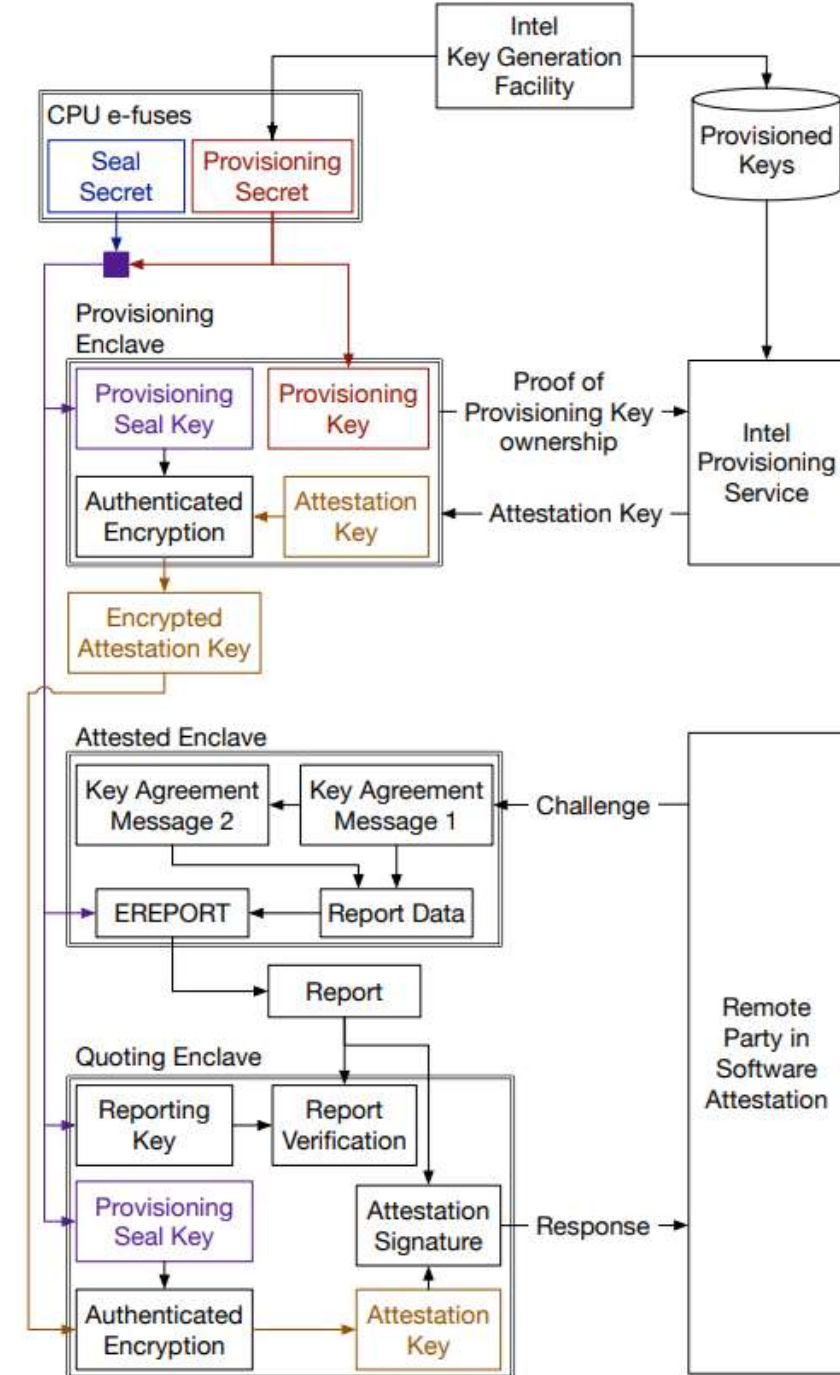


## Confirm Platform

Valid MAC confirms both enclaves execute on the same platform, establishing local trust.

# Remote Attestation

- Remote attestation extends local attestation to prove enclave identity to remote parties.
- The process utilizes a special Quoting Enclave (QE) to verify and transform REPORT structures into signed QUOTE structures.
- Procedure
  - The challenger issues a challenge to the Application Enclave, which generates a manifest containing responses and an ephemeral public key.
  - After creating a REPORT structure, the QE signs it with an EPID private key.
  - The remote server validates the QUOTE signature and verifies manifest integrity.



# Data Sealing Mechanisms

- Sealing to Enclave Identity
  - This policy produces a sealing key available exclusively to instances of a specific enclave, ensuring data can only be unsealed by the exact same enclave code.
- Sealing to Sealing Identity
  - This policy generates a key available to all enclaves signed by a particular sealing authority, enabling data sharing across related enclaves from the same developer.
- Both policies enable secure persistent storage of trusted data on external disks with confidentiality guarantees, allowing enclaves to securely store and retrieve sensitive information across execution sessions.